

Better Elo

Momentum-Responsive Chess
Rating System

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Openings

01

Problem Identification

- Traditional Elo
- Rating Cavities
- Matchmaking
- Skills Assessment
- Current Research

02

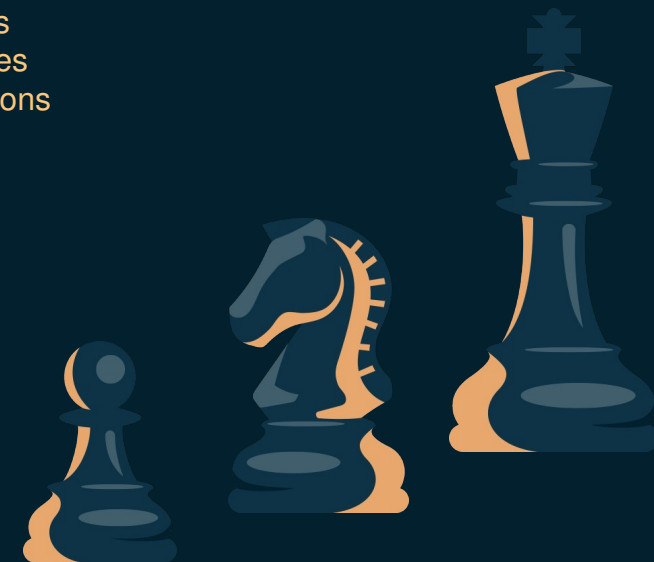
Solution Implementation

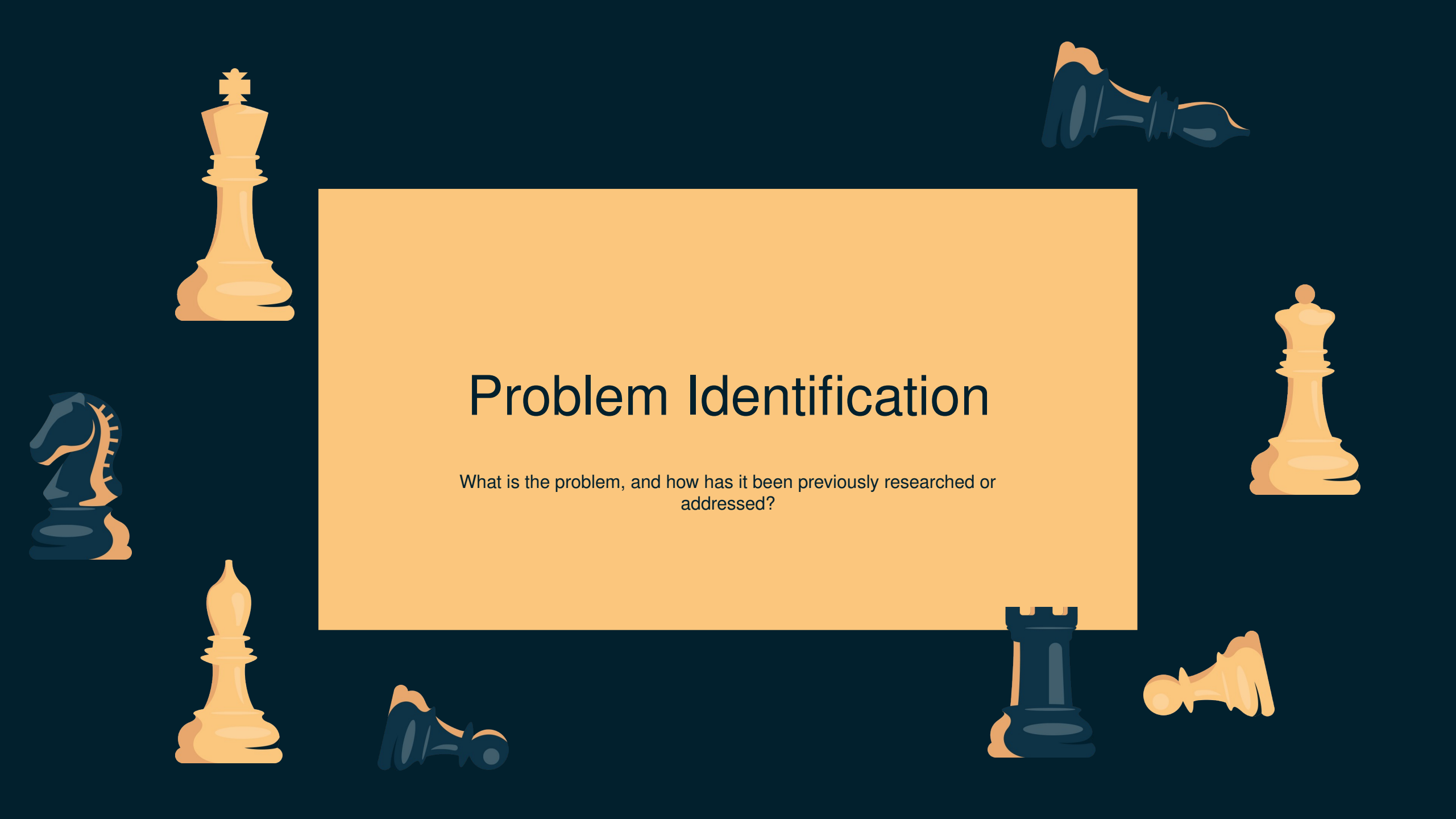
- Goals
- Data Aggregation
- Features
- Evolutionary Algorithm Details
- How Does the EA “fix” the Problem?

03

Results

- Validation
- Statistics
- Outcomes
- Implications



The background is a dark blue gradient. Scattered around a central yellow rectangle are several chess pieces: a white king (top left), a black king (top right), a white king (middle right), a black knight (middle left), a white pawn (bottom left), a black pawn (bottom center), a black rook (bottom right), and a white pawn (bottom right).

Problem Identification

What is the problem, and how has it been previously researched or addressed?

Problem Motivation



Fixed K-Factor

Traditional Elo win prediction uses this equation, where for a majority of a player's time, the `K` value is fixed.

$$P_{win} = \frac{1}{1 + 10^{\frac{elo_{opponent} - elo_{player}}{K^2}}}$$

Rating Cavities and Stagnation

Due to the fixed `K` value, players are prone to prolonged periods where ratings diverge significantly from actual performance, which I coined as a “rating cavity”.

Matchmaking, Assessment & Slow Improvement

In turn, these cavities cause matchmaking and assessment of true players' skills to become less accurate of their short term changes, as Elo focuses more on long-term rating. This can also lead to a slow improvement of chess ranking.

The key is...

Momentum



Current Research



Current Studies in Momentum

There has been multiple studies on the momentum factor in predicting outcomes of games in areas such as basketball, tennis, and e-sports. Most of which were successful in proving statistical significance in using momentum in addition to other factors for better accuracy.



The Blunders of Other Systems

Glicko and Glicko-2 systems have been created to allow newer players to more quickly traverse the rating ladder than traditional Elo. However, this does not support players who are “settled” but improving at a rapid rate.

TrueSkill was developed to use Bayesian inference to model player skill as a distribution to improve upon Glicko 2, but it is extremely expensive to run with minimal impact for chess.

Both systems were overall focused on long-term skill.

The Goal



01

Enhance, not replace, traditional Elo, by improving the accuracy of the prediction of the outcome of the match, an algorithm used by every online and offline chess platform

02

Use differential evolutions to optimize momentum factors across different players



Solution Implementation

How was the solution implemented to fix this mitigate the identified problem?





Data Collection

Players

I collected data from 7 high-profile chess players: MagnusCarlsen, Hikaru, GothamChess, AnishGiri, FabianoCaruana, AnnaCramling, and WesleySo.

History

The data spanned over 2 years of rapid (3 minute game) play time, totaling over 20,000 games in total for the 7 players. This game data was polled through the Chess.com API.

Chronological Ordering

Ensured that I kept the data in chronological order, very important for trends in the momentum factors over the playing window as well as temporal windows, which we will discuss later

“Playing Window” is a term that will be used for the rest of the presentation, this is just a sliding window of time that we observe statistics over. For my algorithm, I used 50 games, but it could be tuned to anything else.



Differential Evolution

(DE/rand/1/bin)

Population: 500

Generation: 10,000 max, w/ early stopping

Fitness: Prediction Improvement over
Traditional Elo



Mutation

Differential Mutation. For each individual, selects 3 random distinct parents x_1 , x_2 , x_3 . Creates mutant vector:

$$v_i = x_{r1} + F \times (x_{r2} - x_{r3})$$

where F is scaling factor (0.3 increasing to 0.9, self-adapting). This mutation strategy implicitly uses a form of parent selection in its $x_1/x_2/x_3$ selection.

Crossover

Binomial Crossover. Combines mutant and parent vectors with a crossover rate (0.9 decreasing to 0.5, self-adapting) which creates the “trial” genome, preserving good components while allowing diversity.

Survivor Selection

Greedy Selection + Hall Of Fame. Each parent is replaced by its trial if and only if the trial has better fitness. Elitism via a Hall Of Fame that preserves the top 5% of individuals and replaces the worst population members with them.

DEs are very good for noisy and non-linear optimizations because it is population based. It is especially useful when it requires continuous parameter adaptation. Mutation and crossover rates being adaptive also helps balance exploitation vs. exploration.

01

Win Streaks

Tracks the hot and cold patterns of a player in the playing window

02

Recent Win Rate

Short term win rate over the playing window

03

Accuracy

Move quality via Stockfish analysis and evaluation of the centipawn loss

The Genome

Represented as floating values

04

Rating Trend

Total Elo change, captures direction and magnitude without considering pace

05

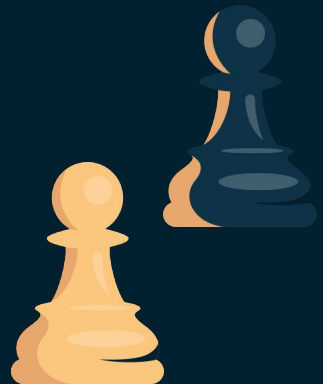
Velocity

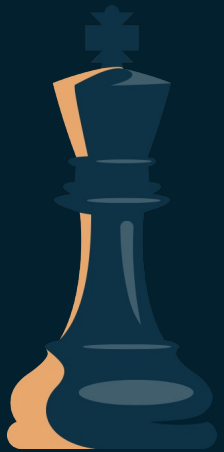
Elo change per game over the playing window, emphasizing speed and consistency

06

Activity

Amount of games in the last 30 days





Again, we are only looking to enhance Elo predictions, so we will use the genome to adjust the Elo probability of winning, which is what is used for matchmaking and similar features:

$$P_{enhanced} = P_{win} + m$$

Our adjustment `m` then becomes:

$$m = \sum_1^6 weight_i * feature_i$$

Using $K = 20$ (most common value) for standard Elo prediction:

$$P_{win} = \frac{1}{1 + 10^{\frac{elo_{opponent} - elo_{player}}{400}}}$$

We properly adjust the probability of the player winning!

Note: the adjustment is clamped to ± 0.2 to ensure that no overwhelming adjustments are made, and the final enhanced probability is clamped to $[0.01, 0.99]$ for no out-of-bounds predictions.

Results

How did the implemented solution perform?



Statistics

Accuracy

Pure Elo predictions were 69.4% accurate, while the momentum-based solution was 72.6% better on average.

+3.2%

Brier Score

Determines how well a probabilistic prediction matches the true prediction. Momentum-based is a 0.219, whereas traditional Elo was 0.245.

+10.6%

AUC-ROC

Measures prediction quality by evaluating how well the algorithm discriminates between wins and losses across all possible probability thresholds. Momentum-based scores a 0.78, while Elo scores 0.74.

+5.4%





Cavity Reduction

01

Frequency

Reduced from 0.05 to 0.0008, a 98% reduction

02

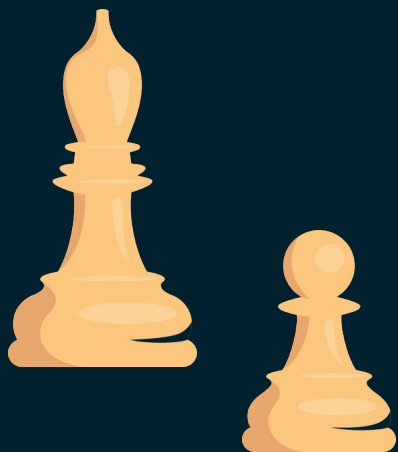
Duration

Reduced from 15.3 games to 1.2 games

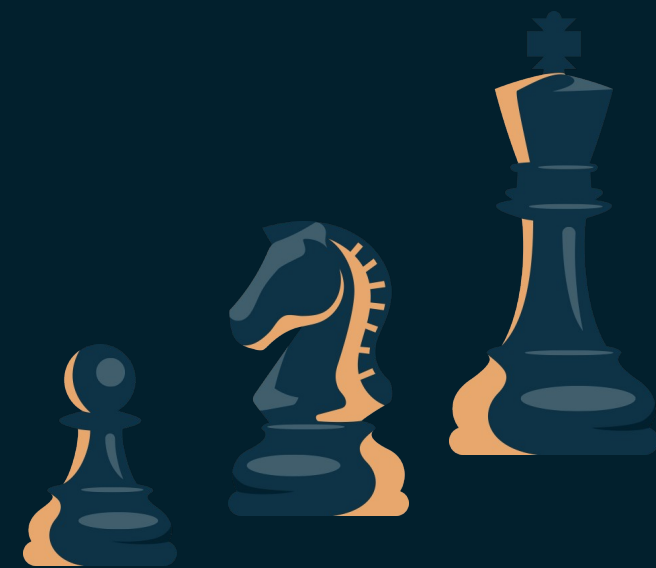
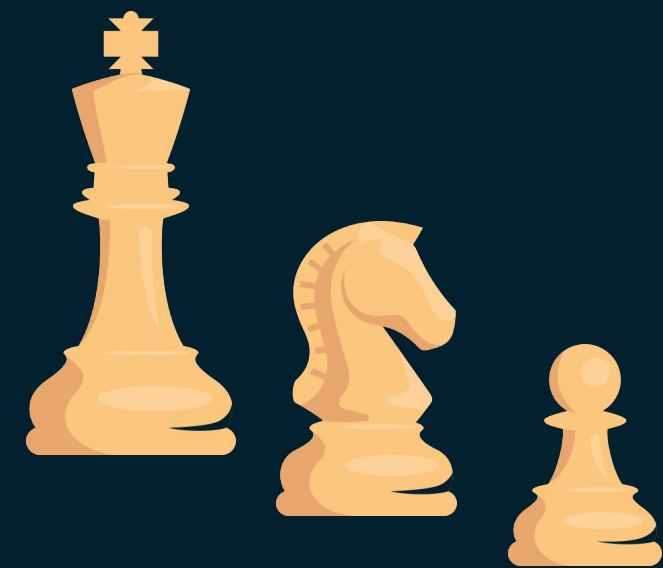
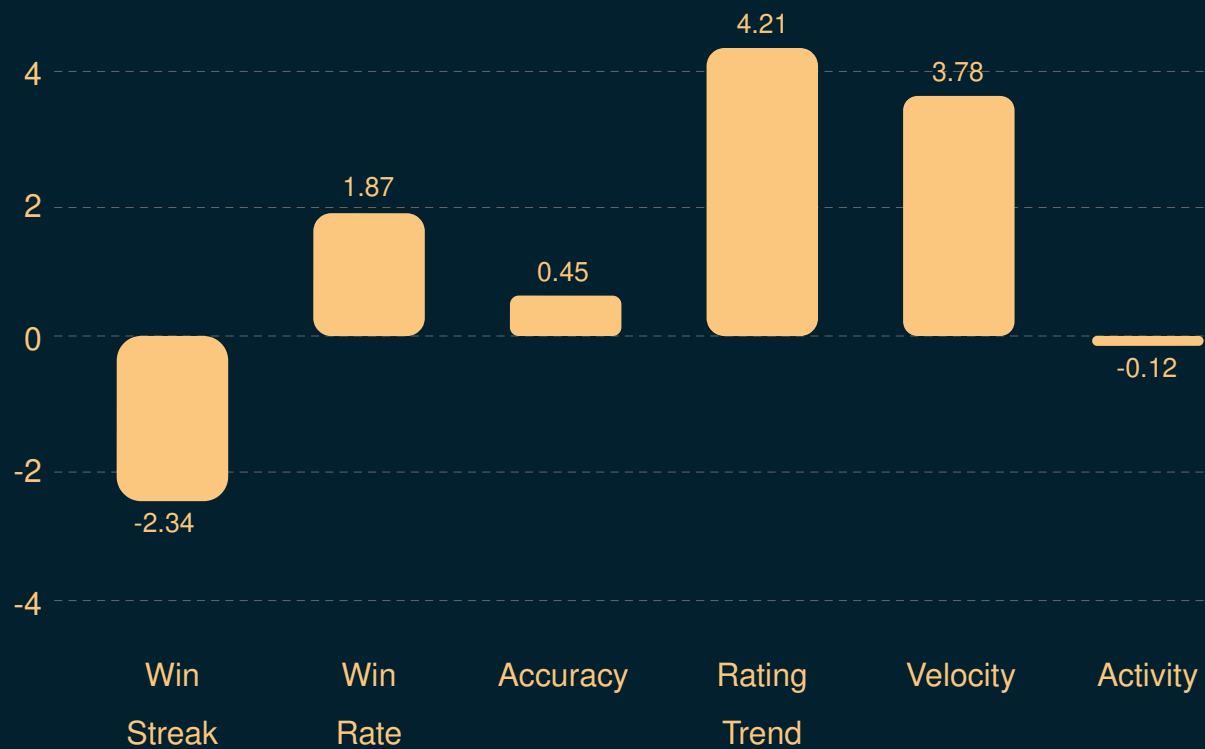
03

Indicators

Much more comprehensive understanding of a user's true skill level



Results (Averages)





Validation

01.

Temporal Validation

See the first 80% of the data, once the algorithm finds the optimal weights, use the remaining 20% to test the predictions

02.

Prediction Horizon

Creates a gap between training and test data to prevent data leakage, ensuring momentum-enhanced Elo predictions are evaluated on genuinely future outcomes for reliability

03.

5-Fold Validation

Run the algorithm 5 times to get some averages and ensure they find near-same values for the weights

04.

McNemar's Test

Evaluates if one prediction method is really better than another when both are tested on the same games



Conclusions and Future Work



Conclusion

Momentum has a meaningful enhancement to the base Elo prediction without sacrifice, and evolutionary optimizations help adapt to patterns and players. Overall, we checkmate the cavity and prediction problem.



Future Work

- Expand to other time constraints
- Investigate other potential statistics to use in the genome
- Real-Time System implementation

Questions?

Want to connect? Scan the QR code below to check out my website,
which has my contact info!

Code available at <https://github.com/DerekCorniello/better-elo>



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